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2
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4
5  Cohort A
6
7  #Question 1
8  #Create Database
9  create database moviedb;
10
11 #Use database
12 use moviedb;
13 Result:      #    Time      Action  Message Duration / Fetch
14 3    1    11:59:14    use moviedb 0 row(s) affected    0.016 sec
15
16 #Create Tables
17 #Membership Table
18 create table MEMBERSHIP(
19     MEM_NUM int,
20     MEM_FNAME varchar(20),
21     MEM_LNAME varchar(20),
22     MEM_STREET varchar(30),
23     MEM_CITY varchar(50),
24     MEM_STATE varchar(30),
25     MEM_ZIP varchar(20),
26     MEM_BALANCE double,
27     primary key(MEM_NUM)
28
29 )
30
31 describe membership;
32
33 Result:
34 MEM_NUM int NO PRI
35 MEM_FNAME varchar(20) YES
36 MEM_LNAME varchar(20) YES
37 MEM_STREET varchar(30) YES
38 MEM_CITY varchar(50) YES
39 MEM_STATE varchar(30) YES
40 MEM_ZIP varchar(20) YES
41 MEM_BALANCE double YES
42
43
44 #Rental Table
45
46 create table RENTAL(
47     MEMBER_NUM int,
48     RENT_NUM int,
49     RENT_DATE date,
50     foreign key(MEMBER_NUM) references MEMBERSHIP(MEM_NUM),
51     primary key(RENT_NUM)
52
53 )
54
55 describe rental;
56 Result:
57
58 MEMBER_NUM int YES MUL
59 RENT_NUM int NO PRI
60 RENT_DATE date YES
61
62
63 #Price Table
64 create table price (
65     PRICE_CODE int,

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66     PRICE_DESCRIPTION varchar(50),
67     PRICE_RENTFEE double,
68     PRICE_DAILYLATEFEE double,
69     primary key(PRICE_CODE)
70 )
71
72 describe price;
73 Result:
74 PRICE_CODE  int NO PRI
75 PRICE_DESCRIPTION  varchar(50) YES
76 PRICE_RENTFEE  double YES
77 PRICE_DAILYLATEFEE  double YES
78
79 #Movie Table
80 create table movie(
81     MOVIE_NUM int,
82     MOVIE_TITLE varchar(30),
83     MOVIE_YEAR int,
84     MOVIE_COST double,
85     MOVIE_GENRE varchar(25),
86     MOVIE_PRICE int,
87     primary key(MOVIE_NUM),
88     foreign key(MOVIE_PRICE) references price(PRICE_CODE)
89 )
90 describe movie;
91 Result:
92 MOVIE_NUM  int NO PRI
93 MOVIE_TITLE  varchar(30) YES
94 MOVIE_YEAR  int YES
95 MOVIE_COST  double YES
96 MOVIE_GENRE  varchar(25) YES
97 MOVIE_PRICE  int YES MUL
98
99 #Video Table
100 create table video(
101     MOVIE_NUMBER int,
102     VID_NUM int,
103     VID_INDATE date,
104     primary key(VID_NUM),
105     foreign key(MOVIE_NUMBER) references movie(MOVIE_NUM)
106 )
107 describe video;
108 Result:
109 MOVIE_NUMBER  int YES MUL
110 VID_NUM int NO PRI
111 VID_INDATE  date YES
112
113
114 #DetailRental
115 create table DetailRental (
116     RENT_NUMBER int,
117     VIDEO_NUMBER int,
118     DETAIL_FEE double,
119     DETAIL_DUEDATE date,
120     DETAIL_RETURNDATE date,
121     DETAIL_DAILYLATEFEE double,
122     foreign key(RENT_NUMBER) references rental(RENT_NUM) ,
123     foreign key(VIDEO_NUMBER) references video(VID_NUM)
124 )
125
126 describe DetailRental;
127
128 Result:
129 RENT_NUMBER int YES MUL
130 VIDEO_NUMBER  int YES MUL

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131 DETAIL_FEE double YES
132 DETAIL_DUEDATE date YES
133 DETAIL_RETURNDATE date YES
134 DETAIL_DAILYLATEFEE double YES
135
136
137 #Question2
138 #Inserting into membership
139 insert into membership
140 values
141 (102,"Tami","Dawson",'2632 Takli Circle',"Norene","TN","37136",11),
142 (103,"Curt","Knight","4025 Cornell Court ","Flatgap","KY ","41219 ",6),
143 (104, "Jamal ", "Melendez", " 788 East 145th Avenue", " Quebeck ", "TN", "38579 ", 0) ,
144 (105 , "Iva", " Mcclain", " 6045 Musket Ball Circle ", "Summit", " KY ", "42783 ", 15 ),
145 (106, " Miranda", " Parks ", "4469 Maxwell Place", " Germantown", " TN", "38183", 0 ),
146 (107, " Rosario ", "Elliott ", "7578 Danner Avenue", " Columbia", " TN", "38402", 5 ),
147 (108, " Mattie", " Guy", " 4390 Evergreen Street", " Lily ", "KY ", "40740", 0 ),
148 (109, " Clint ", "Ochoa ", "1711 Elm Street ", "Greeneville", " TN ", "37745 ", 10 ),
149 (110, " Lewis ", "Rosales", " 4524 Southwind Circle", " Counce ", "TN", "38326", 0 ),
150 (111, " Stacy ", "Mann", " 2789 East Cook Avenue ", "Murfreesboro", " TN", "37132", 8 ),
151 (112, " Luis", " Trujillo", " 7267 Melvin Avenue ", "Heiskell ", "TN", "37754", 3 ),
152 (113 , "Minnie ", "Gonzales ", "6430 Vasili Drive", " Williston", " TN", "38076", 0 );
153
154
155 select * from membership;
156 Result:
157 102 Tami Dawson 2632 Takli Circle Norene TN 37136 11
158 103 Curt Knight 4025 Cornell Court Flatgap KY 41219 6
159 104 Jamal Melendez 788 East 145th Avenue Quebeck TN 38579 0
160 105 Iva Mcclain 6045 Musket Ball Circle Summit KY 42783 15
161 106 Miranda Parks 4469 Maxwell Place Germantown TN 38183 0
162 107 Rosario Elliott 7578 Danner Avenue Columbia TN 38402 5
163 108 Mattie Guy 4390 Evergreen Street Lily KY 40740 0
164 109 Clint Ochoa 1711 Elm Street Greeneville TN 37745 10
165 110 Lewis Rosales 4524 Southwind Circle Counce TN 38326 0
166 111 Stacy Mann 2789 East Cook Avenue Murfreesboro TN 37132 8
167 112 Luis Trujillo 7267 Melvin Avenue Heiskell TN 37754 3
168 113 Minnie Gonzales 6430 Vasili Drive Williston TN 38076 0
169
170
171 #inserting into rental
172 insert into rental
173 values
174 (1001,'2016-03-01',103 ),
175 (1002,'2016-03-01',105 ),
176 (1003,'2016-03-02',102 ),
177 (1004,'2016-03-02',110 ),
178 (1005,'2016-03-02',111 ),
179 (1006,'2016-03-02',107 ),
180 (1007,'2016-03-02',104 ),
181 (1008,'2016-03-03',105 ),
182 (1009,'2016-03-03',111 );
183
184 select * from rental;
185 Result:
186 103 1001 2016-03-01
187 105 1002 2016-03-01
188 102 1003 2016-03-02
189 110 1004 2016-03-02
190 111 1005 2016-03-02
191 107 1006 2016-03-02
192 104 1007 2016-03-02
193 105 1008 2016-03-03
194 111 1009 2016-03-03
195

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196
197 #inserting into price
198 insert into price
199 values
200 (1,"Standard",2 ,1 ),
201 (2,"New Release", 3.5 ,3 ),
202 (3,"Discount",1.5, 1 ),
203 (4 ,"Weekly Special", 1, .5 );
204
205 select * from price;
206 Result:
207 1   Standard      2    1
208 2   New Release  3.5  3
209 3   Discount     1.5  1
210 4   Weekly Special 1    0.5
211
212 #inserting into movie
213 insert into movie
214 values
215 (1234 ,"The Cesar Family Christmas", 2014, 39.95,"FAMILY",2 ),
216 (1235,"Smokey Mountain Wildlife", 2011, 59.95,"ACTION", 1 ),
217 (1236,"Richard Goodhope",2015 ,59.95 ,"DRAMA", 2 ),
218 (1237,"Beatnik Fever", 2014 ,29.95 ,"COMEDY", 2 ),
219 (1238,"Constant Companion", 2015, 89.95 ,"DRAMA",null ),
220 (1239,"Where Hope Dies", 2005 ,25.49 ,"DRAMA", 3 ),
221 (1245,"Time to Burn",2012,45.49,"ACTION", 1 ),
222 (1246,"What He Doesn't Know",2013 ,58.29,"COMEDY", 1 );
223
224 select * from movie;
225 Result:
226 1234   The Cesar Family Christmas   2014    39.95   FAMILY   2
227 1235    Smokey Mountain Wildlife     2011    59.95   ACTION   1
228 1236    Richard Goodhope             2015    59.95   DRAMA    2
229 1237    Beatnik Fever                 2014    29.95   COMEDY   2
230 1238    Constant Companion            2015    89.95   DRAMA
231 1239    Where Hope Dies               2005    25.49   DRAMA    3
232 1245    Time to Burn                  2010    45.49   ACTION   1
233 1246    What He Doesn't Know          2013    58.29   COMEDY   1
234
235
236 #inserting into video
237 insert into video
238 values
239 (54321,'2015-06-18',1234),
240 (54324,'2015-06-18',1234),
241 (54325,'2015-06-18',1234),
242 (34341,'2014-01-22',1235),
243 (34342,'2014-01-22',1235),
244 (34366,'2016-03-02',1236),
245 (34367,'2016-03-02',1236),
246 (34368,'2016-03-02',1236),
247 (34369,'2016-03-02',1236),
248 (44392,'2015-10-21',1237),
249 (44397,'2015-10-21',1237),
250 (59237,'2016-02-14',1237),
251 (61388,'2014-01-25',1239),
252 (61353,'2013-01-28',1245),
253 (61354,'2013-01-28',1245),
254 (61367,'2015-07-30',1246),
255 (61369,'2015-07-30',1246);
256
257 select * from video;
258 Result:
259 1235      34341      2014-01-22
260 1235      34342      2014-01-22

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261 1236      34366      2016-03-02
262 1236      34367      2016-03-02
263 1236      34368      2016-03-02
264 1236      34369      2016-03-02
265 1237      44392      2015-10-21
266 1237      44397      2015-10-21
267 1234      54321      2015-06-18
268 1234      54324      2015-06-18
269 1234      54325      2015-06-18
270 1237      59237      2016-02-14
271 1245      61353      2013-01-28
272 1245      61354      2013-01-28
273 1246      61367      2015-07-30
274 1246      61369      2015-07-30
275 1239      61388      2014-01-25
276
277
278
279 #inserting into DetailRental
280 insert into DetailRental
281 values
282 (1001,34342,2,'2016-03-04','2016-03-02',null),
283 (1001,61353,2,'2016-03-04','2016-03-03',1),
284 (1002,59237,3.5,'2016-03-04','2016-03-04',3),
285 (1003,54325,3.5,'2016-03-04','2016-03-09',3),
286 (1003,61369,2,'2016-03-06','2016-03-09',1),
287 (1003,61388,0,'2016-03-06','2016-03-09',1),
288 (1004,44392,3.5,'2016-03-05','2016-03-07',3),
289 (1004,34367,3.5,'2016-03-05','2016-03-07',3),
290 (1004,34341,2,'2016-03-07','2016-03-07',1),
291 (1005,34342,2,'2016-03-07','2016-03-05',1),
292 (1005,44397,3.5,'2016-03-05','2016-03-05',3),
293 (1006,34366,3.5,'2016-03-05','2016-03-04',3),
294 (1006,61367,2,'2016-03-07',null,1),
295 (1007,34368,3.5,'2016-03-05',null,3),
296 (1008,34369,3.5,'2016-03-05','2016-03-05',3),
297 (1009,54324,3.5,'2016-03-05',null,3),
298 (1001,34366,3.5,'2016-03-04','2016-03-02',3);
299
300 select * from DetailRental;
301
302 Result:
303 1001      34342      2      2016-03-04      2016-03-02
304 1001      61353      2      2016-03-04      2016-03-03      1
305 1002      59237      3.5    2016-03-04      2016-03-04      3
306 1003      54325      3.5    2016-03-04      2016-03-09      3
307 1003      61369      2      2016-03-06      2016-03-09      1
308 1003      61388      0      2016-03-06      2016-03-09      1
309 1004      44392      3.5    2016-03-05      2016-03-07      3
310 1004      34367      3.5    2016-03-05      2016-03-07      3
311 1004      34341      2      2016-03-07      2016-03-07      1
312 1005      34342      2      2016-03-07      2016-03-05      1
313 1005      44397      3.5    2016-03-05      2016-03-05      3
314 1006      34366      3.5    2016-03-05      2016-03-04      3
315 1006      61367      2      2016-03-07                      1
316 1007      34368      3.5    2016-03-05                      3
317 1008      34369      3.5    2016-03-05      2016-03-05      3
318 1009      54324      3.5    2016-03-05                      3
319 1001      34366      3.5    2016-03-04      2016-03-02      3
320
321
322
323 #Question3
324
325 select movie_year

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326 from movie
327 where movie_num = 1245;
328
329 Result Before Change:
330 2012
331
332 update movie
333 set MOVIE_YEAR = 2010
334 where MOVIE_NUM = 1245;
335
336 select movie_year
337 from movie
338 where movie_num=1245;
339
340 Result After Change:
341 2010
342
343
344 #Question4
345
346 select movie_price
347 from movie
348 where movie_genre = "ACTION";
349
350 Result Before Change:
351 1
352 1
353
354 update movie
355 set movie_price =3
356 where movie_genre ="ACTION";
357
358 select movie_price
359 from movie
360 where movie_genre = "ACTION";
361
362 Result After Change:
363 3
364 3
365
366 #Question 5
367 select price_rentfee
368 from price;
369
370
371 Result Before Change:
372 2
373 3.5
374 1.5
375 1
376
377 update price
378 set price_rentfee = price_rentfee+0.50;
379
380 select price_rentfee
381 from price;
382
383 Result After Change:
384 2.5
385 4
386 2
387 1.5
388
389
390 #Question6
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391 select movie_title,movie_year,movie_genre
392 from movie;
393
394 Result:
395 The Cesar Family Christmas    2014    FAMILY
396 Smokey Mountain Wildlife      2011    ACTION
397 Richard Goodhope              2015    DRAMA
398 Beatnik Fever                 2014    COMEDY
399 Constant Companion            2015    DRAMA
400 Where Hope Dies               2005    DRAMA
401 Time to Burn                  2010    ACTION
402 What He Doesn't Know          2013    COMEDY
403
404 #Question7
405
406 select movie_year,movie_title,movie_cost
407 from movie
408 order by movie_year desc;
409
410 Result:
411 2015    Richard Goodhope    59.95
412 2015    Constant Companion  89.95
413 2014    The Cesar Family Christmas  39.95
414 2014    Beatnik Fever      29.95
415 2013    What He Doesn't Know  58.29
416 2011    Smokey Mountain Wildlife  59.95
417 2010    Time to Burn        45.49
418 2005    Where Hope Dies     25.49
419
420 #Question 8
421
422 select movie_title,movie_year,movie_genre
423 from movie
424 order by movie_genre asc,movie_year desc;
425
426 Result:
427 Smokey Mountain Wildlife      2011    ACTION
428 Time to Burn                  2010    ACTION
429 Beatnik Fever                 2014    COMEDY
430 What He Doesn't Know          2013    COMEDY
431 Richard Goodhope              2015    DRAMA
432 Constant Companion            2015    DRAMA
433 Where Hope Dies               2005    DRAMA
434 The Cesar Family Christmas    2014    FAMILY
435
436 #Question9
437
438 select movie_num,movie_title,movie_price
439 from movie
440 where movie_title like 'R%';
441
442 Result:
443 1236    Richard Goodhope      2
444
445 #Question 10
446 select movie_title,movie_year,movie_cost
447 from movie
448 where movie_title like '%hope%'
449 order by movie_title asc;
450
451 Result:
452 Richard Goodhope      2015    59.95
453 Where Hope Dies       2005    25.49
454
455 #Question 11

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456 select movie_title,movie_year,movie_genre
457 from movie
458 where movie_genre = 'ACTION';
459
460 Result:
461 Smokey Mountain Wildlife      2011    ACTION
462 Time to Burn                  2010    ACTION
463
464
465 #Question 12
466 select movie_num,movie_title,movie_cost
467 from movie
468 where movie_cost > 40;
469
470 Result:
471 1235    Smokey Mountain Wildlife    59.95
472 1236    Richard Goodhope            59.95
473 1238    Constant Companion          89.95
474 1245    Time to Burn                45.49
475 1246    What He Doesn't Know        58.29
476
477
478 #Question 13
479 select movie_num,movie_title,movie_cost,movie_genre
480 from movie
481 where movie_genre in ('ACTION','COMEDY') and movie_cost <50
482 order by movie_genre asc;
483
484
485 Result:
486 1245    Time to Burn      45.49    ACTION
487 1237    Beatnik Fever    29.95    COMEDY
488
489 #Question 14
490 select mem_num,mem_fname,mem_lname,mem_street,mem_state,mem_balance
491 from membership
492 where mem_state = 'TN'
493    and mem_balance < 5
494    and mem_street like '%Avenue';
495
496 Result:
497 104  Jamal   Melendez  788 East 145th Avenue  TN  0
498 112  Luis    Trujillo  7267 Melvin Avenue    TN  3
499
500 #Question 15
501 select movie_genre, count(*) as Number_Of_Movies
502 from movie
503 group by movie_genre;
504
505 Result:
506 FAMILY  1
507 ACTION  2
508 DRAMA   3
509 COMEDY  2
510
511
512 #Question 16
513
514 select round(avg(movie_cost),4) as Average_Movie_Cost
515 from movie;
516
517 Result:
518 51.1275
519
520 #Question17

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521 select movie_genre, round(avg(movie_cost),2) as Average_Cost
522 from movie
523 group by movie_genre;
524
525 Result:
526 FAMILY  39.95
527 ACTION  52.72
528 DRAMA   58.46
529 COMEDY  44.12
530
531
532 #Question 18
533 select movie_title,movie_genre,price_description,price_rentfee
534 from movie
535 join price
536 on movie.movie_price=price.price_code;
537
538 Result:
539 What He Doesn't Know    COMEDY  Standard    2.5
540 The Cesar Family Christmas  FAMILY  New Release  4
541 Richard Goodhope        DRAMA    New Release  4
542 Beatnik Fever            COMEDY  New Release  4
543 Smokey Mountain Wildlife  ACTION  Discount    2
544 Where Hope Dies          DRAMA    Discount    2
545 Time to Burn             ACTION  Discount    2
546
547
548 #Question19
549 select movie_genre,round(avg(price_rentfee),2) as Average_Rental_Fee
550 from movie
551 join price
552 on movie.movie_price=price.price_code
553 group by movie_genre;
554
555
556 Result:
557 COMEDY  3.25
558 FAMILY  4
559 DRAMA   3
560 ACTION  2
561
562
563 #Question20
564 select movie_title, round((movie_cost/price_rentfee),2) as breakeven_rentals
565 from movie
566 join price
567 on movie.movie_price=price.price_code
568 where movie_price is not null;
569
570 Result:
571 What He Doesn't Know    23.32
572 The Cesar Family Christmas  9.99
573 Richard Goodhope        14.99
574 Beatnik Fever           7.49
575 Smokey Mountain Wildlife  29.98
576 Where Hope Dies         12.74
577 Time to Burn            22.74
578
579 #Question21
580 select movie_title,movie_year
581 from movie
582 where movie_price is not null;
583
584 Result:
585 What He Doesn't Know    2013

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586 The Cesar Family Christmas 2014
587 Richard Goodhope 2015
588 Beatnik Fever 2014
589 Smokey Mountain Wildlife 2011
590 Where Hope Dies 2005
591 Time to Burn 2010
592
593
594 #Question22
595 select movie_title,movie_genre,movie_cost
596 from movie
597 where movie_cost between 44.99 and 49.99;
598
599 Result:
600 Time to Burn ACTION 45.49
601
602 #Question23
603 select movie_title,price_description,price_rentfee,movie_genre
604 from movie
605 join price
606 on movie.movie_price=price.price_code
607 where movie_genre in ('Family','Comedy','Drama');
608
609 Result:
610 The Cesar Family Christmas New Release 4 FAMILY
611 Richard Goodhope New Release 4 DRAMA
612 Beatnik Fever New Release 4 COMEDY
613 Where Hope Dies Discount 2 DRAMA
614 What He Doesn't Know Standard 2.5 COMEDY
615
616
617 #Question24
618 select distinct mem_num,mem_fname,mem_lname,mem_balance
619 from membership
620 join rental
621 on membership.mem_num=rental.member_num;
622
623
624 Result:
625 102 Tami Dawson 11
626 103 Curt Knight 6
627 104 Jamal Melendez 0
628 105 Iva Mcclain 15
629 107 Rosario Elliott 5
630 110 Lewis Rosales 0
631 111 Stacy Mann 8
632
633 #Question25
634 select min(mem_balance) as Minimum_Balance,max(mem_balance) as
Maximum_Balance,round(avg(mem_balance),2) as Average_Balance
635 from membership
636 join rental
637 on membership.mem_num=rental.member_num;
638
639 Result:
640 0 15 7.56
641
642 #Question26
643 select rent_number,rent_date,video_number,movie_title,detail_duedate,detail_returndate
644 from DetailRental
645 join rental
646 on rental.rent_num = DetailRental.rent_number
647 join video
648 on video.vid_num = DetailRental.video_number
649 join movie

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650 on movie.movie_num = video.movie_number
651 where detail_returndate > detail_duedate
652 order by rent_number ,movie_title;
653
654 Result:
655 1003      2016-03-02   54325   The Cesar Family Christmas  2016-03-04   2016-03-09
656 1003      2016-03-02   61369   What He Doesn't Know      2016-03-06   2016-03-09
657 1003      2016-03-02   61388   Where Hope Dies  2016-03-06   2016-03-09
658 1004      2016-03-02   44392   Beatnik Fever    2016-03-05   2016-03-07
659 1004      2016-03-02   34367   Richard Goodhope  2016-03-05   2016-03-07
660
661 #Question27
662 select rent_number,rent_date,movie_title,detail_fee
663 from DetailRental
664 join rental
665 on rental.rent_num = DetailRental.rent_number
666 join video
667 on video.vid_num = DetailRental.video_number
668 join movie
669 on movie.movie_num = video.movie_number
670 where detail_returndate <= detail_duedate
671 order by movie_title;
672
673 Result:
674 1002      2016-03-01   Beatnik Fever    3.5
675 1005      2016-03-02   Beatnik Fever    3.5
676 1006      2016-03-02   Richard Goodhope  3.5
677 1008      2016-03-03   Richard Goodhope  3.5
678 1001      2016-03-01   Richard Goodhope  3.5
679 1001      2016-03-01   Smokey Mountain Wildlife  2
680 1004      2016-03-02   Smokey Mountain Wildlife  2
681 1005      2016-03-02   Smokey Mountain Wildlife  2
682 1001      2016-03-01   Time to Burn     2
683
684 #Question28
685 select movie_num, movie_genre, movie_cost,
686        round(avg(movie_cost) over (partition by movie_genre),2) as Average_Cost,
687        round(((movie_cost - avg(movie_cost) over (partition by movie_genre)) /
688        avg(movie_cost) over (partition by movie_genre)) * 100, 2) as Percent_Difference
689 from movie
690 order by movie_genre, movie_num;
691
692 Result:
693 1235      ACTION   59.95    52.72    13.71
694 1245      ACTION   45.49    52.72   -13.71
695 1237      COMEDY   29.95    44.12   -32.12
696 1246      COMEDY   58.29    44.12    32.12
697 1236      DRAMA    59.95    58.46     2.54
698 1238      DRAMA    89.95    58.46    53.86
699 1239      DRAMA    25.49    58.46   -56.4
700 1234      FAMILY   39.95    39.95     0

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